

# Seonhong Min

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## RESEARCH INTERESTS

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Lattice-based Cryptography, especially Fully Homomorphic Encryption (FHE).

## EDUCATION

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**Seoul National University** Mar. 2022 – Present  
Integrated M.S./Ph.D. in Computer Science & Engineering  
Advisor: Prof. Yongsoo Song

**Seoul National University** Mar. 2018 - Feb. 2022  
B.S. in Mathematics

**Daegu Science High School** Mar. 2015 – Feb. 2018

## PUBLICATIONS

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Authors are listed in alphabetical order by last name, unless an asterisk(\*) is indicated.

## CONFERENCES

C10. 2026/322

**Multi-key Fully Homomorphic Encryption with Non-Interactive Setup in the Plain Model**  
*Seonhong Min, Jeongeun Park, Yongsoo Song*  
IACR CRYPTO 2026

C9. 2025/1804

**\*HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping**  
*Binwu Xiang, Seonhong Min, Intak Hwang, Zhiwei Wang, Haoqi He, Yuanju Wei, Yi Deng, Jiang Zhang, Kang Yang, Yu Yu*  
IACR EUROCRYPT 2026

C8. 2025/382

**On the Security and Privacy of CKKS-based Homomorphic Evaluation Protocols**  
*Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
IACR ASIACRYPT 2025

C7. 2024/2032

**Carousel: Fully Homomorphic Encryption from Slot Blind Rotation Technique**  
*Intak Hwang, Seonhong Min, Yongsoo Song*  
IACR ASIACRYPT 2025

C6. 2025/216

**Practical (Malicious) Circuit Privacy / Sanitization for TFHE**

*Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
ACM CCS 2025

C5. 2025/1255

**Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition**

*Intak Hwang, Shinwon Lee, Seonhong Min, Yongsoo Song*  
SAC 2025

C4. 2024/1502

**MatriGear: Accelerated Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing**

*Hyunho Cha, Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
IEEE S&P 2025

C3. 2025/429

**\*Enhanced CKKS Bootstrapping with Generalized Polynomial Composites Approximation**

*Seonhong Min, Joon-woo Lee, Yongsoo Song*  
ACM AsiaCCS 2025

C2. 2022/1460

**Towards Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity**

*Hyesun Kwak, Seonhong Min, Yongsoo Song*  
IACR PKC 2024

C1. 2023/958

**Faster TFHE Bootstrapping with Block Binary Keys**

*Changmin Lee, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
ACM AsiaCCS 2023

**JOURNALS**

J2. 2024/1534

**More Efficient Lattice-based OLE from Circuit-private Linear HE with Polynomial Overhead**

*Leo de Castro, Duhyeon Kim, Miran Kim, Keewoo Lee, Seonhong Min, Yongsoo Song*  
ACR Communications in Cryptology

J1. 2406.14372

**\*Ring-LWE based encrypted controller with unlimited number of recursive multiplications and effect of error growth**

*Yeongjun Jang, Joowon Lee, Seonhong Min, Hyesun Kwak, Junsoo Kim, Yongsoo Song*  
IEEE Trans. on Control of Network Systems

**PREPRINTS**

P3. 2025/2057

**\*Distributed Key Generation for Efficient Threshold-CKKS**

*Seonhong Min, Guillaume Hanrot, Jai Hyun Park, Alain Passelègue, Damien Stehlé*

P2. 2025/203

**Ciphertext-Simulatable HE from BFV with Randomized Evaluation**

*Intak Hwang, Seonhong Min, Yongsoo Song*

P1. 2024/181

**Functional Bootstrapping for Packed Ciphertexts via Homomorphic LUT Evaluation**

*Dongwon Lee, Seonhong Min, Yongsoo Song*

## PRESENTATION

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**Enhanced CKKS Bootstrapping with Generalized Polynomial Composites Approximation**

AsiaCCS 2025

Youtube

**Faster TFHE Bootstrapping with Block Binary Keys**

FHE.org Meetup

**Faster TFHE Bootstrapping with Block Binary Keys**

AsiaCCS 2023

Youtube

**Functional Bootstrapping for Packed Ciphertexts via Homomorphic LUT Evaluation**

FHE.org Meetup

Youtube

**Towards Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity**

PKC 2024

## POSTERS

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**Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity**

FHE.org 2023

**Carousel: Blind Rotation Over the Automorphism Group**

FHE.org 2024

**Practical Sanitization for TFHE**

FHE.org 2024

**MatriGear: Accelerated Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing**

FHE.org 2024

## HONORS & AWARDS

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**Korea Cryptography Contest 2025**

3rd Place, 4th Place

**Korea Cryptography Contest 2024**  
3rd Place

**Korea Cryptography Contest 2023**  
4th Place

## EXPERIENCES

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**FHElab Lyon (Cryptolab Inc.)** (Intern)

2024.11 – 2025.02

## GITHUB REPOSITORIES

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**Multi-key TFHE**

<https://github.com/SNUCP/MKTFHE>

**Carousel.jl**

<https://github.com/SNUCP/Carousel.jl>

**HIENAA.jl**

<https://github.com/snu-lukemin/HIENAA.jl>

**Ciphertext Simulatable BFV**

<https://github.com/SNUCP/simct>

**CRS-less MKFHE**

<https://github.com/SNUCP/crsless>

## ACADEMIC SERVICES

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**Reviewer at**

TCS (Theoretical Computer Science), DCC (Designs, Codes and Cryptography)

## SKILLS

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**Languages**

Korean (native), English (fluent), Japanese (conversational)

**Programming Languages**

Julia, Python, Java, Go,  $\text{\LaTeX}$